# Your Brain Is Made of Cells KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** 86 billion axon cell body dendrites fires synapse

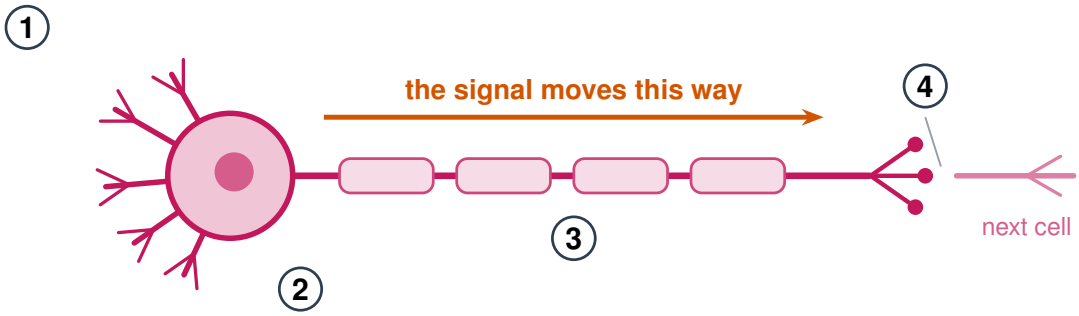
*You can use a word more than once.*

**1. Catch the ruler.** Write where you caught it. Use the table to find your time.

| **Caught at** | 5 cm | 10 cm | 15 cm | 20 cm | 25 cm | 30 cm |
|---|---|---|---|---|---|---|
| **Time** | 101 ms | 143 ms | 175 ms | 202 ms | 226 ms | 247 ms |

| | **Try 1** | **Try 2** | **Try 3** | **Best time** |
|---|---|---|---|---|
| **Caught at (cm)** | Answers vary. | | | |
| **Time (ms)** | | | | |

**2. The parts of a neuron.**




**1** = dendrites **2** = cell body

**3** = axon **4** = synapse

The dendrites take signals in. The cell body adds them up.

If the total is big enough, the neuron fires. The axon carries the signal away.

At the synapse, chemicals carry the signal to the next neuron.

Your brain has about 86 billion neurons.

**3. The squeeze chain.**

| | Try 1 | Try 2 | Try 3 |
|---|---|---|---|
| **Total time (seconds)** | Answers vary. | | |

Number of people in the circle: _____ (If you went around 5 times, multiply by 5.)

Time for one person = $\frac{\text{total time}}{\text{number of people}}$ = Answers vary. About 0.2 to 0.4 s.

**4. Talk about it.** One neuron is slow. How can a brain be fast?

A brain can be fast because billions of neurons work at the same time.

**Go Further (Day 2).** The click test. Write your 5 times in milliseconds.

_____ _____ _____ _____ _____ Average: _____

Is your click time close to your ruler time? Answers vary. Usually yes: 200 to 300 ms.